

What does it mean?

Discussion, honest limitations, and a Conclusion that lands

YOU LEAVE WITH

A Discussion plan built from three or four 'moves', and a Conclusion whose first sentence answers your question.



Tutorial 7 of 8 • Semester 1

Today's plan

Goal: Leave with a Discussion plan and a Conclusion that answers your question.

- 1 Methods: lags, causation and mechanisms
- 2 Results vs Discussion; a menu of Discussion moves
- 3 Limitations that build trust
- 4 Conclusions that land, and the mirror test

BRING / HAVE READY

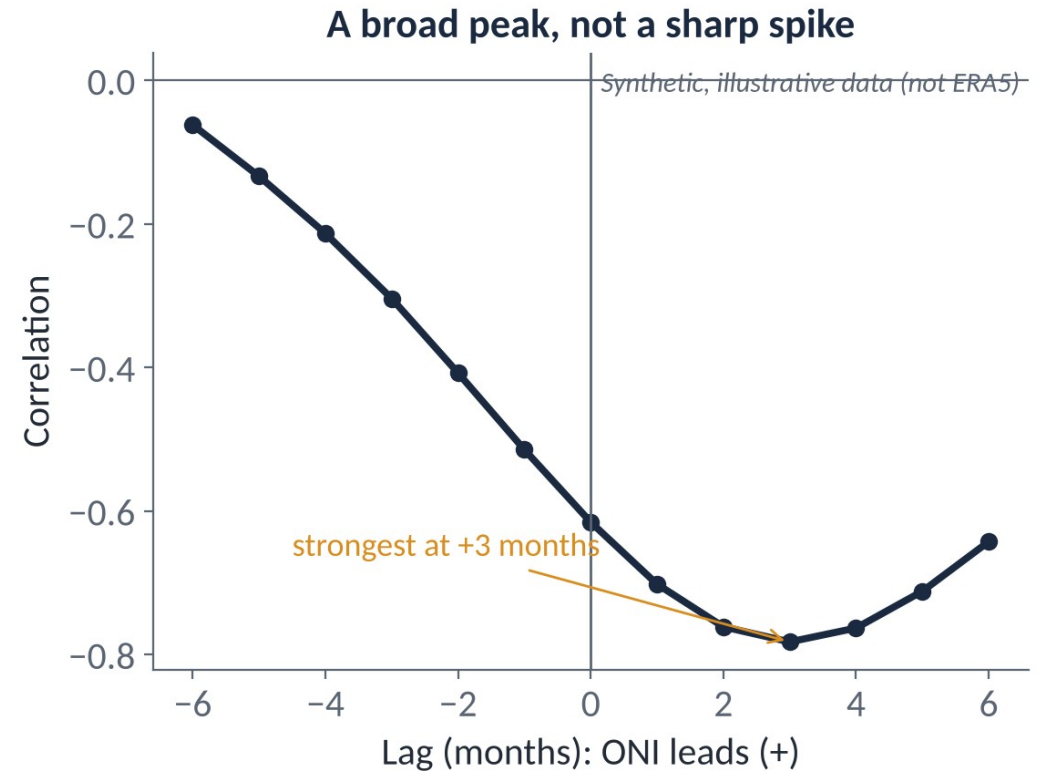
- Revised Results (6A) and Introduction (6B)
- Task 6C Discussion brainstorm
- Your 'Discussion parking lot' from last week



METHODS

Lags, timing and causation

- **Lag correlation:** shift the ONI against cloud and recompute r . In notebook §6, positive lag = **ONI leads**.
- Peaks are **broad** because both series have long memory: neighbouring lags give similar r .
- A peak at +3 months is **consistent with** a delayed response, but doesn't prove ENSO causes it, and the ONI is already a 3-month mean.
- From correlation to cause you need a **mechanism**, a **time order**, and **independent evidence** (another variable).
- Until then write 'is associated with' or 'is consistent with', not 'causes'.



METHODS

From correlation to mechanism: cloud is a symptom

- The notebook can load other ERA5 fields: change VARIABLE to **skt** (skin temperature $\approx$ SST over the ocean), **tcwv** (water vapour), **tclw** / **tciw** (cloud liquid / ice).
- **A consistent story across variables is strong evidence:** warmer SST + more water vapour + more high cloud over the central Pacific during El Niño $\rightarrow$ convection shifted east.
- **Low cloud:** warmer SST under a stratocumulus deck tends to weaken the inversion and thin the cloud, but stability isn't in this dataset, so say so.
- ENSO events differ in **where** the warming is strongest (central vs eastern Pacific 'flavours'). Composites blend them.
- One figure from another variable can earn its place, **if** it tests the mechanism.

TRY IT (5 MIN IN THE NOTEBOOK)

Set VARIABLE = "skt" and rerun §2

- Does the El Niño – La Niña SST pattern line up with your cloud signal?
- In the real data, skin temperature over Niño 4 has $r \approx +0.9$ with the ONI.
- If your region's cloud signal has **no** SST signal under it, what else could drive it?

Results say what; Discussion says why and so what

- **Results:** what pattern, sign, magnitude, relationship? (evidence)
- **Discussion:** what does it mean physically, how sure are we, and how does it fit what others found? (interpretation)
- Schultz: separate speculation explicitly, e.g. 'We speculate that...'. It avoids reviewers concluding you can't tell results from speculation.
- Don't **re-describe** results in the Discussion; refer back briefly ("the weaker MC relationship (Section 3)") and **interpret**.

USEFUL PHRASES

- This is consistent with...
- One explanation is...; an alternative is...
- This differs from [source], possibly because...
- We speculate that...
- This result does not hold for..., which suggests...
- If this relationship holds, the 2026–27 El Niño would be expected to...

A menu of Discussion moves: choose three or four

EXPLAIN

Give a physical mechanism for the main result: Walker circulation, subsidence, stability, SST, convection.

COMPARE

Agree or disagree with published work? If you disagree, say why that might be.

EXCEPTIONS

Discuss events, regions or seasons that don't fit. They are often the most interesting part.

ALTERNATIVES

Could something other than your preferred explanation produce the same pattern?

LIMITATIONS

The ones that genuinely affect your answer, with their likely effect.

BACK TO THE QUESTION

Return to the Introduction's question: did you answer it? What remains open?

~450 words is room for 3–4 moves done well. **EXPLAIN** and **BACK TO THE QUESTION** are essential; pick the others that fit your result.

Limitations that build trust

Schultz: being forthright about limitations ‘will enhance, not reduce, your credibility’. Astute readers will spot them anyway.

WEAK

“There are some limitations to this study, such as the short dataset. More data would improve the results and further research is needed.”

STRONG

“The 54 El Niño months come from only about six events, and 14 of them belong to the exceptionally strong 2015–16 event. Recomputing the composite without 2015–16 would show whether the CP signal is a property of El Niño in general or of one extreme event. In addition, ERA5 cloud cover comes from the model's cloud scheme rather than direct observation, so the magnitude (though probably not the sign) of the response should be treated with caution.”

WHAT CHANGED

- **Specific:** says exactly what is limited, how much, and its likely **effect on the answer**.
- Suggests a **concrete check** instead of ‘more research is needed’.
- Picks limitations that **matter** for this question, not a generic list.

Conclusions that land

Schultz: answer the question; two or three take-home findings; nothing new; end ‘forcefully, not on a whimper’.

A WHIMPER

“In conclusion, ENSO affects clouds in many different ways and it is a complicated system. More research is needed to understand this further.”

A LANDING

“High cloud responds to ENSO in opposite senses on either side of the Pacific warm pool: over CP it increases by about [X] percentage points during El Niño, while over MC it decreases by about [Y]. The MC response is weaker and less consistent between events, so a single ‘ENSO cloud response’ is misleading. Compositing central- and eastern-Pacific El Niño events separately would test whether event type explains the inconsistency.” (illustrative)

WHAT CHANGED

- **Sentence 1 answers the question** in the same words and tags as the purpose statement.
- Findings carry their **key number**. No new results, figures or citations.
- ‘Further research’ becomes a **specific, testable next step**, or is left out.

The mirror test

END OF INTRODUCTION

“The purpose of this report is to test whether high cloud over the central Pacific (CP) and the Maritime Continent (MC) responds to ENSO in opposite senses, and to compare the strength of each relationship with the ONI.”

START OF CONCLUSION

“High cloud over CP and MC responds to ENSO in opposite senses, and the MC relationship with the ONI is the weaker of the two ($r = []$ vs $[]$).”

PUT THEM SIDE BY SIDE AND ASK

- Does the conclusion answer **exactly** what the purpose promised: not more, not less?
- Are the **same tag-phrases** used (CP, MC, ONI)?
- Would a reader who read only these two boxes know your question **and** your answer? (Many will read only these.)

IN-CLASS ACTIVITY • 16 MIN

Plan the Discussion, draft the landing

- 1 From your parking lot and brainstorm, choose **3–4 moves** from the menu. EXPLAIN and BACK TO THE QUESTION are compulsory.
- 2 Write the **topic sentence** for each Discussion paragraph.
- 3 For your best limitation, write it the ‘strong’ way: **what, how much, effect on the answer, a check**.
- 4 Draft the **first sentence of your Conclusion**: it must answer your question.
- 5 With a partner, run the **mirror test** on your purpose statement and that sentence.

YOU LEAVE WITH

- Discussion topic sentences + one strong limitation
- Conclusion sentence 1, mirror-tested
- Everything you need to finish the draft by next week

PAIR CHECK

- Is the EXPLAIN paragraph a **mechanism**, or just ‘ENSO affects...’?
- Is the limitation tied to its **effect on the answer**?
- Mirror test: same question, same tags?

Before Tutorial 8 • Thu 3 Dec

One week. About 3 hours. Next week you bring a COMPLETE first draft: the natural end of this term's writing.

7A • 75 min

Discussion draft, ~450 words

- Your 3–4 moves, one paragraph each
- Mechanism + honest limitation + back to the question
- Flag speculation as speculation

7B • 25 min

Conclusion, 100–150 words

- Sentence 1 answers the question
- 2–3 findings with key numbers
- No new material; a strong last sentence

7C • 60 min

Assemble the full draft

- Title • all five sections • captions • references
- Word count ($\leq$ ~2,200 is fine for now)
- **Bring it printed or as a PDF on 3 Dec**

NOTEBOOK STRAND

- **Freeze the analysis.** Polish figures only (labels, units, legibility).
- Make sure every number in the text matches the figures.

READ • SCHULTZ (2009) ELOQUENT SCIENCE

- §4.9–4.11 Results, Discussion, Conclusion
- §7.2 Making writing accessible (limitations)